

SLIVEN, Bulgaria

WMO: 156400

Lat: 42.667N

Lon: 26.333E

Elev: 257

StdP: 98.28

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-5.4	-13.8	1.2	-6.0	-11.1	1.5	-4.2	23.8	-0.8	19.6	-0.9	4.7	320	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	33.6	23.0	32.0	22.4	30.5	21.7	24.6	30.9	23.5	30.0	22.5	29.0	3.1	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.7	18.0	27.8	21.5	16.7	27.3	20.3	15.4	26.6	75.7	30.8	71.5	30.1	67.6	29.0	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.0	(b) 14.0	(c) 9.6	(e) -10.1	(f) 36.6	(g) 2.3	(h) 2.3	(i) -11.8	(j) 38.2	(k) -13.1	(l) 39.5	(m) -14.4	(n) 40.8	(o) -16.1	(p) 42.5		
(a) 18.0	(b) 14.0	(c) 9.6	(e) -10.9	(f) 25.5	(g) 2.3	(h) 1.7	(i) -12.6	(j) 26.7	(k) -14.0	(l) 27.7	(m) -15.3	(n) 28.6	(o) -17.0	(p) 29.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	2.2	4.3	7.8	12.5	17.4	21.7	24.3	24.4	19.7	13.9	8.7	3.7
	DBStd	8.57	4.27	4.55	4.04	3.55	3.27	3.00	2.55	2.71	3.37	3.86	4.18	4.22
	HDD10.0	798	242	164	91	16	1	0	0	0	0	11	73	201
	HDD18.3	2371	499	393	326	178	56	6	1	0	26	142	290	455
	CDD10.0	2047	1	4	24	90	231	350	442	445	289	133	33	4
	CDD18.3	578	0	0	0	2	28	106	184	187	65	6	0	0
	CDH23.3	5012	0	0	1	18	188	860	1704	1755	449	37	0	0
	CDH26.7	1792	0	0	0	1	23	265	666	710	124	3	0	0
Wind	WSAvg	2.8	2.6	2.9	3.2	2.6	2.7	2.9	3.3	3.0	2.9	2.7	2.3	2.9
Precipitation	PrecAvg	587	42	37	41	45	66	63	47	39	56	47	49	53
	PrecMax	1051	101	115	82	90	186	194	139	81	169	140	132	133
	PrecMin	299	2	1	1	4	21	4	2	3	2	2	1	0
	PrecStd	148	27	28	23	20	38	43	35	25	47	33	29	37
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.3	17.3	20.9	25.9	28.9	33.5	36.4	35.9	32.2	26.7	21.2	16.1
		MCWB	10.2	11.7	14.0	18.6	20.0	23.3	24.0	22.9	22.1	19.0	15.9	12.2
	2%	DB	11.7	15.0	18.6	22.6	26.9	31.3	33.7	33.7	29.9	24.0	18.4	13.5
		MCWB	8.6	10.3	12.4	16.1	19.0	22.4	23.1	23.0	20.8	17.6	14.6	10.9
	5%	DB	9.9	12.9	16.4	20.4	25.5	29.5	31.8	32.1	27.9	21.8	16.5	11.5
		MCWB	7.5	9.1	11.3	14.3	18.4	21.5	22.3	22.5	19.9	16.4	13.4	9.5
	10%	DB	8.3	10.8	14.5	18.7	23.9	28.0	30.3	30.5	25.9	19.8	14.6	9.8
		MCWB	6.4	7.8	10.0	13.3	17.5	20.5	21.5	21.7	18.9	15.3	12.1	8.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	12.8	14.8	19.4	21.6	25.5	25.8	26.0	23.8	20.4	17.3	13.9
		MCDB	13.5	15.6	19.6	24.7	26.6	30.4	32.8	32.6	29.7	25.0	19.5	15.1
	2%	WB	9.2	11.0	13.1	16.9	20.1	23.9	24.3	24.5	22.0	18.3	15.1	11.3
		MCDB	11.1	14.1	17.4	21.4	25.6	28.9	31.3	31.2	27.8	22.4	17.7	12.9
	5%	WB	7.9	9.4	11.7	15.1	19.0	22.5	23.2	23.4	20.7	16.9	13.8	9.7
		MCDB	9.5	12.4	15.7	19.4	24.2	28.1	30.0	30.3	26.6	21.0	16.0	11.1
	10%	WB	6.6	8.1	10.4	13.7	17.9	21.2	22.2	22.4	19.4	15.7	12.5	8.2
		MCDB	8.1	10.6	13.9	17.8	23.0	27.0	28.8	29.3	24.8	19.3	14.4	9.9
Mean Daily Temperature Range	5% DB	MDBR	5.8	6.9	8.1	9.1	9.7	9.7	10.0	10.3	9.5	7.9	6.5	5.9
		MCDBR	7.9	10.3	11.7	12.0	12.0	12.2	12.9	12.7	12.6	11.1	8.7	7.6
		MCWBR	5.8	6.8	7.1	6.9	6.5	5.9	6.2	6.1	6.3	6.3	5.7	5.7
	5% WB	MCDBR	7.1	9.4	11.0	11.3	11.2	11.2	11.9	11.8	12.0	10.4	7.7	6.8
		MCWBR	5.5	6.6	7.0	7.0	6.5	6.1	6.2	6.2	6.5	6.5	5.3	5.5
Clear-Sky Solar Irradiance	taub	0.345	0.377	0.421	0.475	0.449	0.451	0.457	0.467	0.439	0.401	0.371	0.347	
	taud	2.306	2.221	2.123	1.982	2.133	2.166	2.163	2.130	2.175	2.263	2.293	2.305	
	Ebn at Noon	773	805	811	794	830	827	817	794	785	768	732	730	
	Edh at Noon	87	111	138	172	152	147	146	146	130	106	88	80	
All-Sky Solar Radiation	RadAvg	1.52	2.30	3.39	4.67	5.83	6.33	6.51	5.88	4.33	2.81	1.73	1.33	
	RadStd	0.21	0.28	0.30	0.50	0.45	0.49	0.45	0.30	0.35	0.29	0.20	0.18	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.51	N/A	N/A	N/A	+1.22	+1.56	-89	-127	N/A	N/A
(47) Regional (30 neighbors)		+0.42	N/A	N/A	N/A	+0.57	+0.58	N/A	-114	N/A	N/A

Nomenclature: See separate page